AudioMind: The Ambient Voice Cockpit

An open-source neural audio player built specifically for developers using modern AI coding environments.



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THE ATTENTION VACUUM

MODERN AGENTIC AI CODING HARNESSES ARE EXTRAORDINARILY POWERFUL, BUT STANDARD WORKFLOWS CREATE SEVERE FRICTION.

DISPATCH COMPLEX REFACTOR TO GOOGLE ANTIGRAVITY

THE 15-MINUTE STARE

AI FINISHED 15 MINUTES AGO. USER ZONED OUT.

COMPLETE

IDLE

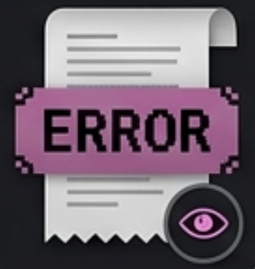
THE IDLE TAX

Developers dispatch tasks and lose 15+ minutes waiting silently, not knowing the AI is ready for the next instruction.



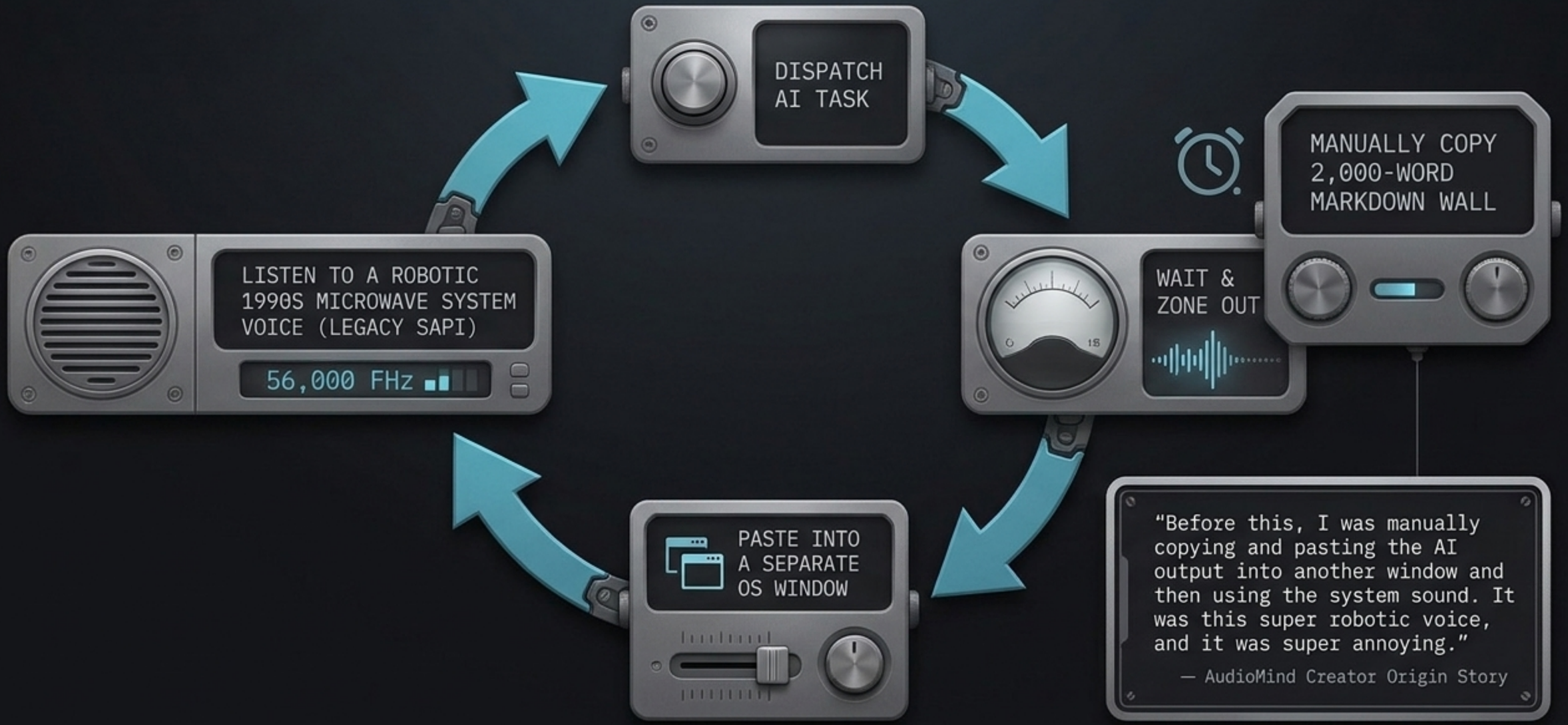
THE READING BOTTLENECK

Staring at 2,000-word markdown walls of code diffs and architectural explanations causes severe eye strain and mental fatigue.



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THE VICIOUS CYCLE OF THE COPY-PASTE HACK



> “Before this, I was manually copying and pasting the AI output into another window and then using the system sound. It was this super robotic voice, and it was super annoying.”
>
> — AudioMind Creator Origin Story

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Zero Friction. Zero Idle Time.

AudioMind turns your AI assistant into an interactive, hands-free audiobook.
No manual copying. No reading bottlenecks. Just ambient flow state.

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The Anti-Distraction Engine

Old Way



AudioMind



Zero-Latency Auto-Play

You never miss a completion. The exact millisecond an AI agent finishes its response, AudioMind chimes and begins streaming. Maintain hands-free ambient multitasking whether you are looking at your screen, reading documentation, or pacing across the room.

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Code as an Audio Album

Listening to code is fundamentally different from listening to a podcast. It requires tactile control. **AudioMind** splits markdown into navigable chapters on the fly.



FLOATING DESKTOP HUD



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Why Standard Voice Assistants Fail Developers

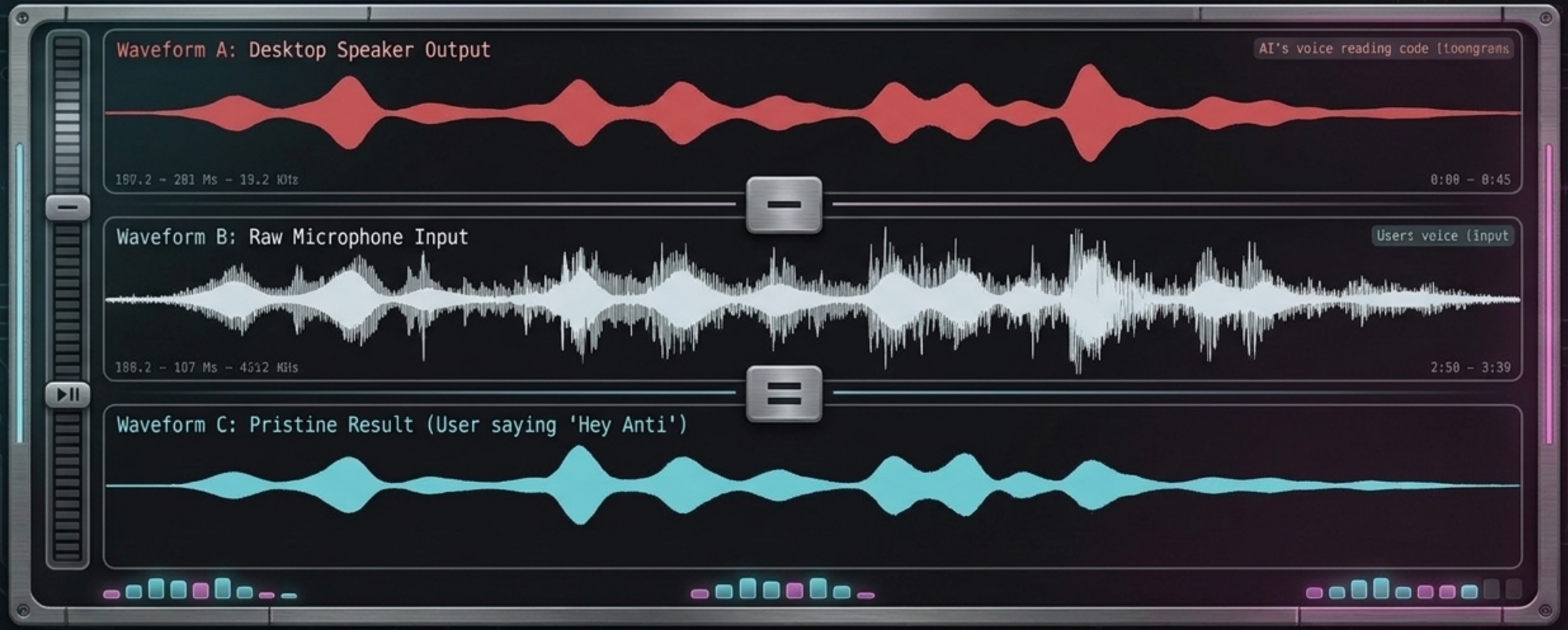
Standard tools are built for basic queries, not dense **architectural explanations**.
AudioMind **is engineered explicitly** for the coding environment.

Features	Smart Speakers	Screen Readers	AudioMind
Latency & Observation	Requires manual trigger	Manual copy-paste	Instant transcript auto-play
Transport Controls	Flat, un-interruptible stream	Basic play/pause	Granular CD-Player HUD (Rewind 5s, Skip Chapter)
IDE Security	Blindly executes commands	Read-only	Zero-Trust AST Firewall
Interruptibility	Mic feedback loop	Read-only	Hardware-level WASAPI AEC

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Hardware-Level Acoustic Echo Cancellation

Interrupt the AI without wearing headphones. AudioMind interfaces directly with the Windows WASAPI loopback and microphone capture streams. Using onboard DSP hardware, it **cleanly subtracts the assistant's's own voice** from the microphone feed. Speak over the **assistant** at **normal conversation** ↳, **levels**, and the on-device wake-word engine hears you instantly.



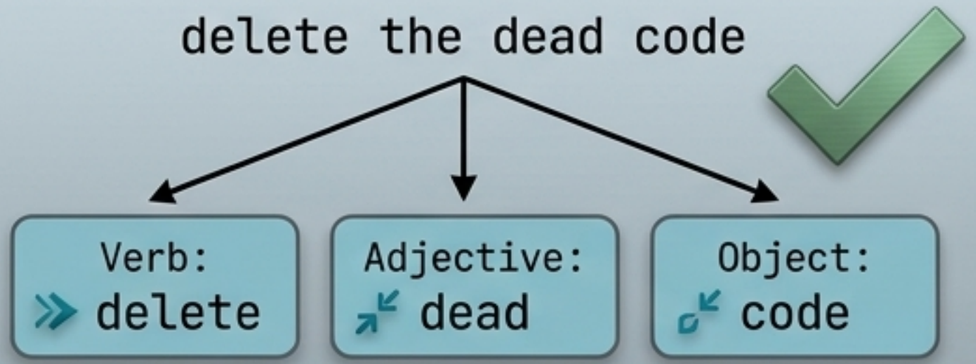
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Semantic Tree / Filter Diagram

Dumb Regex Approach



AST Tokenizer Approach



The Zero-Trust AST Command Firewall

Allowing voice to trigger IDE actions is inherently risky. AudioMind uses a **multi-layer security model**:

AST Tokenizer: Evaluates commands, arguments, and flags with semantic precision.

Win32 UIPI Focus Guard: Verifies targeted IDE windows hold active foreground focus and matching elevation tokens, preventing background keystroke hijacking.

Staged Intent Confirmation: Voice commands paste into the IDE with auto-submit disabled (`press_enter=False`). The human must visually inspect and press Enter.

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Tiered Offline & Neural Speech Architecture

Studio-grade audio. Never locked into a single cloud API. Never stranded without internet.

Tier 1: Local Neural

Kokoro-82M ONNX local neural synthesis
(100% offline, 48kHz stereo, zero cloud latency)

Failover

Tier 2: Cloud Neural

High-definition Microsoft Edge-TTS
neural cloud voices

Circuit-Breaker

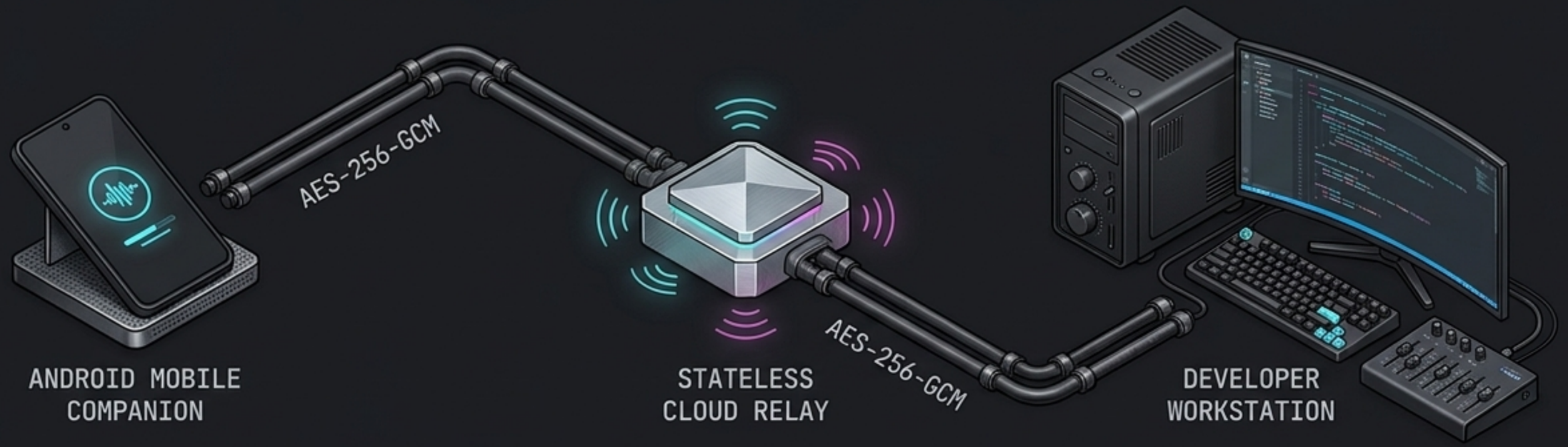
Tier 3: Emergency Fail-Safe

Local Windows SAPI 5.0 fallback

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Unbound by the Desk

The Android Mobile Companion & Cloud Relay

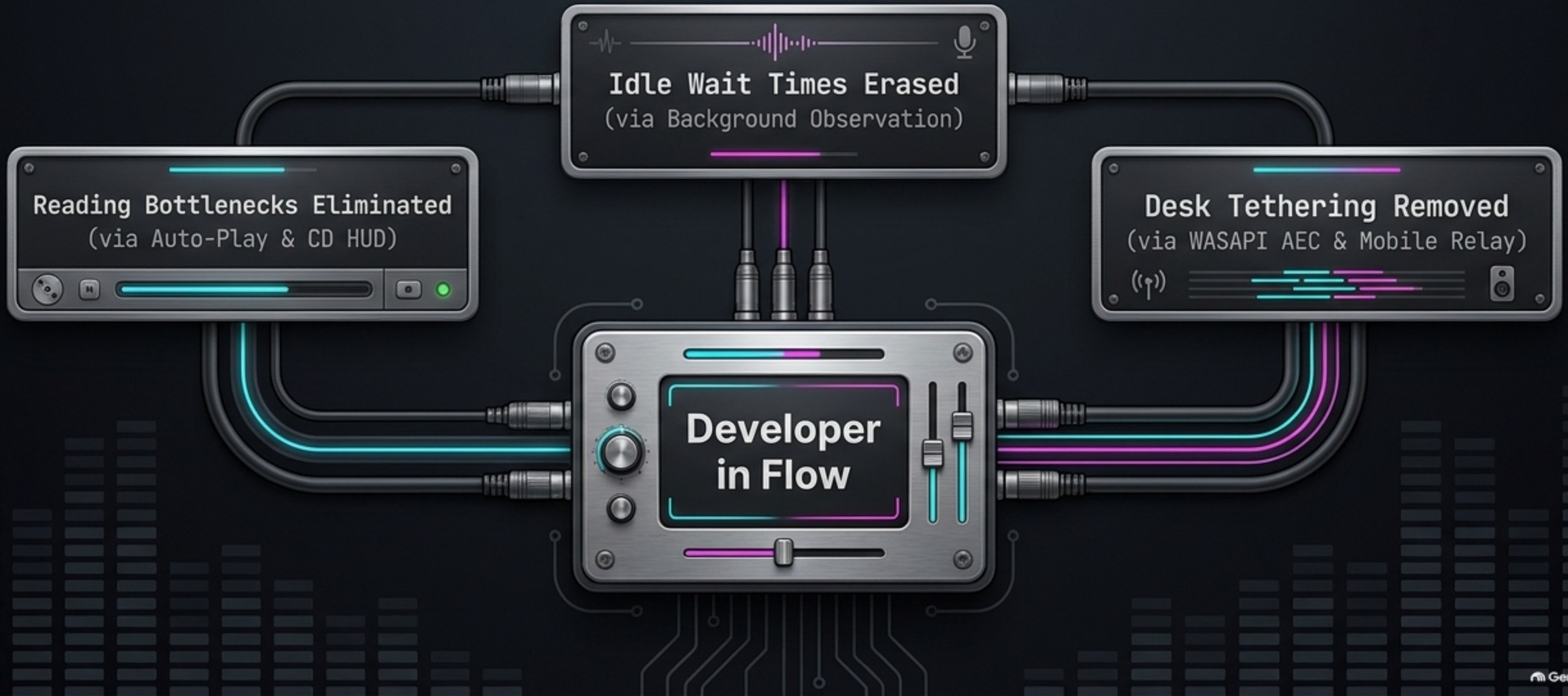


- Trigger **complex refactoring tasks** from across the room using **Google Assistant App Actions**: 'Hey Google, ask Anti to run tests.'
- Secured via **EncryptedSharedPreferences** backed by the **Android Hardware KeyStore**.
- Connects through a **stateless FastAPI cloud relay** secured by **SHA-256 HMAC bearer authentication**.

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The Ambient Flow State

AudioMind is not just a voice assistant. It is a paradigm shift in how we **interact with agentic AI**. By removing the friction of text processing, wait times, and hardware limitations, it creates a seamless ambient multitasking environment.



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The Aha Moments



1. The Copy-Paste Hack Backstory

The frustration that started it all: manually copying code to trigger 1990s robotic OS voices just to avoid screen fatigue.



2. CD-Player Controls for Code

Why a simple 'play/pause' fails. Code is dense. You need granular control to 'rewind 5 seconds' or 'skip chapter' when navigating architectural diffs.



3. AST Security over Dumb Regex

Why standard regex breaks developer workflows (blocking 'rm' or 'model' due to substring matches), and why AST semantic parsing is required for IDEs.



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The Open Source Ethos



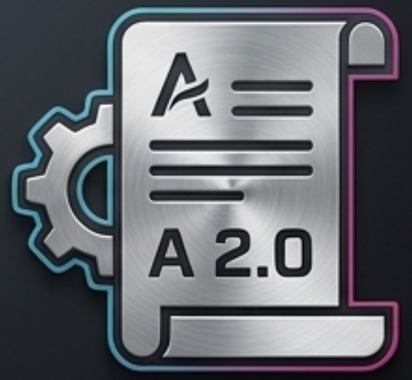
Zero Telemetry

No usage tracking. Your code and prompts stay yours.



100% Offline Capable

Run entirely locally with ONNX models and WASAPI.



Apache 2.0 Licensed

Free, open, and extensible.

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Initialize Your Cockpit



GitHub Repository:	adamm285-dev/audiomind
Supported Harnesses:	Google Antigravity, VS Code, Cursor, Terminal
Platform:	Windows 10/11 (with Android Companion)
License:	Apache License 2.0

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